Harsh Verma

hverma@wpi.edu | linkedin.com/in/harsh-v01 | Portfolio | +1(508)-849-1786 | Worcester, MA

SKILLS

- **Technical:** C++, Python (ML: PyTorch), Git, MATLAB, Robot Operating System (ROS), Linux, OpenCV, Gazebo, Ansys, CFD, CAD/CAM (SolidWorks, Fusion 360, PowerMILL), 3D printing.
- **Soft Skills:** Teamwork, Adaptability, Problem-Solving, Conflict Management.

PROJECTS

Realtime Path Planning in Uncertain Environments with Dynamic Obstacles (C++):

- Implemented RRTx and RT-RRT*, two real-time variants of RRT in C++. Path to goal dynamically changed at each instance.
- Executed three types of samplings, including sampling in Elliptical region like Informed-RRT*.
- Simulated the working of the algorithm with moving obstacles in a static maze environment. Used the OpenFrameworks library in C++ to visualize.

Linear Quadratic Regulator (LQR) for Quadcopter position in real-time:

- Implemented the LQR controller in a quadcopter simulation to intercept a moving UAV in the airspace.
- Determined the State-Space Representation, Cost Function and calculated the LQR gains for the MATLAB simulation.

Manipulation using ROS and OpenManipulator-X arm (C++ and Python):

- Carried out pick-and-place operations using a 4-DOF robotic arm with gripper. Controlled joints using PID controller.
- Used Robot Operating System to integrate various control scripts to control the robot. Calculated the Forward and Inverse Kinematics for the Arm and converted to code.

Computer Vision/Deep Learning- Traffic Sign Classification using on NVIDIA Jetson Nano (Python):

- Developed a Traffic Sign Classification system for cars to read speed limit signs on roads using a Jetson Nano and camera.
- Trained and deployed a Convolutional Neural Network model (AlexNet) completely on Jetson Nano using PyTorch. Used the GTSRB dataset and later developed own dataset for increased accuracy.

Metaheuristic optimization algorithms & Reinforcement Learning to improve Robotic Assembly Line Efficiency:

- Modified nature-inspired optimization algorithms to im the assignment of robots in an assembly line.
- robot assignment to different workstations of industrial robotic assembly lines. Implemented code completely in MATLAB.
- Improved efficiency of the Honey Badger and Aquila optimization algorithms by embedding Q-tables (RL).
- Cut down manufacturing time by 9% by generating solutions with the Aquila optimization algorithm compared to the conventional Genetic Algorithm. Project received top grade 'S' as the final-year Capstone Project in Mechanical Engineering.

EXPERIENCE

Aircraft Design (CAD & CFD) Engineer — **Fort Worth, Texas**

SAE Aero Design East 2022 (Student Team Competition) — 2021-2022

- World **rank 3** in 2022 season. Designed a 3-meter wingspan aircraft and its winglets- wing shape, aspect ratio, Lift-to-Drag ratio. Proposed a unique lift-generating airfoil-shaped fuselage.
- Modelled the complex geometry of the aircraft's fuselage and winglets in SolidWorks (Computer Aided Design). Analyzed the mission scoring equation on MATLAB to maximize the score and found the best payload size.
- Created documentation for the *Technical Design Report* and *presented* the Technical Presentation to the judge Engineers from Lockheed Martin and Boeing.

Machine Design and Operations Intern — **India**

Garware Technical Fibres Ltd. — June 2022 – July 2022

- Reduced product assembly time by 10% - replicated existing rope-twisting machinery in CAD, modified machinery and manufactured the parts using lathe machine.
- Increased Rope Winder capacity by 25% by modifying the Mechanism.

EDUCATION

Worcester Polytechnic Institute (In Progress) — **Worcester, MA**

- *Master of Science:* Robotics Engineering (*GPA: 4.00/4.00*) — 2023 – 2025

Vellore Institute of Technology — **India**

- *Bachelor of Technology:* Mechanical Engineering — 2019 – 2023

AWARDS & CERTIFICATIONS

- **CSWP (SolidWorks)** - Mechanical Design, Sheet Metal, Surfacing, Drawing Tools
- **SAE Aero Design East 2022** - Stood #3 out of 18 international teams. Rank 1 in both Technical Design Report and Technical Presentation. Rank 1 in the Asia-Pacific region.